Removing Torus Artifacts Reveals Nonperturbative Loop Locality in Quantum Spin Ice

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Periodic clusters of quantum spin ice admit noncontractible spin rearrangements that are absent from the infinite lattice. On the 32-site face-centred-cubic cluster (FCC-32), the shortest artifact is a four-spin process at order $J_\pm^2$, one order before the physical hexagon process at $|J_\pm|^3$. We remove it by folding virtual spinon states into the 2970-state ice manifold with the Feshbach map and deleting matrix elements between different covering-lattice transport sectors. Every finite-cluster calculation in this work uses FCC-32. Matched periodic and masked ice-band thermodynamics show that the operation removes the false second-order scale over the full range for which all 2970 ice-descended roots remain resolved. A direct 120-state calculation of $S^{zz}(\boldsymbol{q},\omega)$ at all eight FCC-32 momenta finds periodic inelastic weight 0.313 at $\boldsymbol{q}=0$ for $J_\pm/J_{zz}=-0.20$, whereas the masked value is 1.4×10^{-29} and every nonzero momentum remains active. We then resolve the masked Hamiltonian by simple-loop perimeter L. Leading perturbation theory predicts $K_L^{(0)}=2\binom{L-2}{L/2-1}|J_\pm|^{L/2}/J_{zz}^{L/2-1}$ and rapid loop proliferation. The nonperturbative amplitudes remain strongly ordered: at $J_\pm/J_{zz}=-0.30$, a deeper-virtual-space calculation gives $K_8/K_6=0.262$, $K_{10}/K_6=0.0617$, and $K_{12}/K_6=0.0204$. Thus the six-site ring remains the dominant elementary loop even where bare order counting predicts otherwise.

I. INTRODUCTION

Quantum spin ice (QSI) realizes a three-dimensional compact $U(1)$ gauge theory with gapped gauge charges and an emergent photon [1–3]. Its simplest microscopic derivation restricts the Hilbert space to the classical two-in–two-out manifold and generates a six-spin ring operator around each pyrochlore hexagon. The derivation is controlled at weak transverse exchange. Away from that limit it leaves a quantitative question: does the dynamics remain local in loop perimeter, or do longer gauge processes proliferate?

Exact diagonalization is attractive because it works for both signs of the transverse exchange, including the sign associated with a π-flux background and a QMC sign problem. The relevant finite-size error, however, is topological rather than merely a shortage of sites. A periodic cluster is a torus. A short spin string may close through a periodic image even though the corresponding path is open on the covering lattice. Such a process can appear at lower perturbative order than the physical hexagon and manufacture the dominant low-energy scale.

We address the artifact and the loop physics with one construction on one cluster. The Feshbach map folds virtual spinon excursions into the FCC-32 ice manifold. We resolve the result by the spin transported across the covering lattice and keep only transport-preserving matrix elements. The method is tested three ways: against sign-free QMC thermodynamics, against the expected weak-coupling power, and through the momentum-resolved longitudinal response. The corrected map is then used as an operator microscope. Every surviving off-diagonal element is classified by the geometry of the spin set it flips, and its amplitude is compared with an analytic perturbative ratio.

The resulting picture is simple. Transport masking removes the false torus scale and the associated response at $\boldsymbol{q}=0$. It does not reveal a runaway hierarchy of long loops. Instead, virtual spinon dressing suppresses longer simple loops increasingly strongly with perimeter. Corrections to the hexagon exist, but the six-site ring remains the leading elementary magnetic operator through the strongest coupling studied.

II. MODEL AND THE TORUS ARTIFACT

We study the nearest-neighbour spin-1/2 XXZ model on the pyrochlore lattice,

$$H = J_{zz}\sum_{\langle ij\rangle} S_i^z S_j^z - J_\pm \sum_{\langle ij\rangle}\left(S_i^+S_j^- + S_i^-S_j^+\right), \qquad (1)$$

where $J_{zz}>0$, $J_\pm$ is the transverse exchange, $S_i^\pm = S_i^x \pm iS_i^y$, and $\langle ij\rangle$ denotes a nearest-neighbour bond. Energy and temperature are reported in units of J_{zz}. At $J_\pm=0$, the ground states satisfy $\sum_{i\in t} S_i^z=0$ on every tetrahedron t. Their projector is P; $Q=1-P$ projects onto configurations containing spinon defects.

All numerical results use the $2\times2\times2$ FCC cluster with $N=32$ spins, which we call FCC-32. Its ice manifold contains 2970 states. On the covering lattice, the shortest exchange sequence connecting distinct ice states flips an alternating six-site hexagon and has leading magnitude

$$g_6 = \frac{12|J_\pm|^3}{J_{zz}^2}. \qquad (2)$$

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FCC-32 also has 24 distinct four-spin paths that close only through periodic identification. They produce the false scale

$$g_4 = \frac{4J_\pm^2}{J_{zz}}, \qquad \frac{g_4}{g_6} = \frac{J_{zz}}{3|J_\pm|}. \tag{3}$$

The complete FCC-32 loop census contains 26,304 four-spin ice-to-ice matrix elements; the transport mask below removes all of them. Because g_4/g_6 diverges as $J_\pm \to 0$, the artifact changes the leading power of the low-energy scale rather than supplying a small finite-size shift.

III. TRANSPORT-MASKED FESHBACH MAP

For complex energy z, the exact Feshbach map onto the ice manifold is

$$F(z) = PHP + PHQ\,(z - QHQ)^{-1} QHP. \tag{4}$$

The second term leaves the ice manifold, propagates through virtual spinon states, and returns. It contains the physical hexagon, the winding artifact, and all longer processes generated within the retained space. Without a mask, roots of $E \in \operatorname{spec} F(E)$ reproduce the ice-descended eigenvalues of the periodic microscopic Hamiltonian.

To distinguish the return paths, choose one lift of FCC-32 to its covering lattice. An ice configuration a has polarization coordinate

$$\boldsymbol{p}_a = \sum_{i=1}^{N} S_i^z(a)\boldsymbol{r}_i, \tag{5}$$

where $\boldsymbol{r}_i$ is the lifted position of site i. Its transport label is $\mathcal{T}_a = 2\boldsymbol{p}_a$. The arbitrary origin cancels because every ice state has the same total S^z. A contractible closed rearrangement connects states with $\mathcal{T}_a = \mathcal{T}_b$; a string closed by periodic identification changes the label by a nonzero box vector.

The transport mask Δ is therefore

$$[\Delta F(z)]_{ab} = F_{ab}(z)\,\delta_{\mathcal{T}_a,\mathcal{T}_b}, \tag{6}$$

where a, b label ice configurations and the last factor is a Kronecker delta between their three-component transport labels. The order of operations matters: PHP alone has not generated either virtual process, so we first fold Q and then mask $F(z)$. Self-consistent roots of $\Delta F(z)$ define the corrected finite-volume spectrum. No counterterm or polar-pullback Hamiltonian is used.

Operationally, the construction has only two steps. First, Eq. (4) sums every virtual excursion that leaves and returns to the ice manifold. Second, Eq. (6) asks whether the return also closes on the covering lattice. If it does, the matrix element is retained; if it closes only after periodic identification, it is deleted. The criterion is geometric and has no fitted coefficient.

The full FCC-32 complement is too large to retain. We define the *virtual depth* d as the maximum number of transverse exchanges needed to reach a retained configuration from any ice state. Thermodynamics and dynamical response use $d = 1$, a common 140,442-state microscopic space for periodic and masked calculations. Loop tomography uses the complete $d = 2$ space of 2,511,450 configurations for the coupling scan. At the strongest coupling we test convergence with the same 16 deterministic ice columns at $d = 1, 2, 3, 4$, retaining every simple-loop partner of those columns in the full 2970-state ice manifold. Each depth is evaluated at the ground energy of its own truncated microscopic Hamiltonian. Depth d is therefore a numerical convergence parameter; it is unrelated to the physical loop perimeter L defined below.

IV. FCC-32 BENCHMARKS

A. Matched ice-band thermodynamics

At each coupling we evaluate the expensive Q-space resolvent at one shared set of energy anchors. Diagonalizing $F(z)$ gives the naive periodic branches; diagonalizing $\Delta F(z)$ gives the masked branches. Monotone interpolation of each branch determines its self-consistent root. The two spectra therefore share the same host, virtual space, anchor grid, and 2970-state count.

For either spectrum $\{E_n\}$, the ice-band heat capacity per spin is

$$\frac{C}{N} = \frac{\langle E^2\rangle_T - \langle E\rangle_T^2}{NT^2}, \tag{7}$$

where T is temperature and $\langle\cdot\rangle_T$ is the canonical average over the 2970 roots. Equation (7) deliberately describes low-energy band thermodynamics. It does not include the extensive high-temperature spinon entropy and is not presented as a full 2^{32} spectrum.

Figure 1(a) is an external sign-free benchmark at $J_\pm/J_{zz} = +0.046$. The comparison is restricted to the low-temperature ice feature, where the 2970-state band is the appropriate sector. The QMC peak lies at $T/J_{zz} = 1.10 \times 10^{-3}$. The naive periodic FCC-32 peak is 6.59×10^{-3}, whereas masking moves it to 5.72×10^{-4}. Thus the false winding process shifts the apparent scale upward by a factor of six; removing it brings the peak within a factor of two of QMC. We do not resolve the residual finite-volume and virtual-depth offset here.

The coupling scan in Fig. 1(b) then tests the method without external input. For each spectrum we track the heat-capacity maximum with the largest peak height; this remains unambiguous when the periodic curve develops a smaller secondary feature. Periodic and masked peaks otherwise use the same extraction rule. The periodic branch follows the lower-order torus scale, while the mask restores the contractible-loop scale and remains defined into the nonperturbative regime. A log–log fit to the first three points gives powers 1.80 (peri-

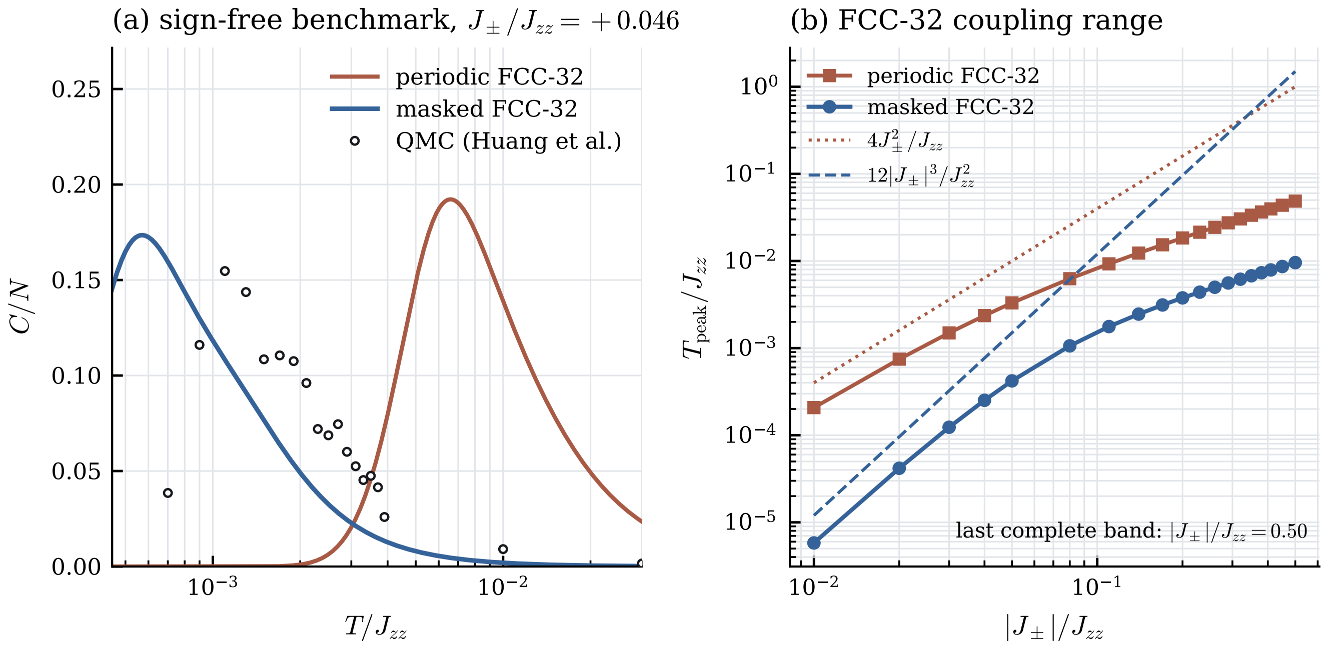

FIG. 1. FCC-32 ice-band thermodynamics. (a) Periodic and transport-masked heat capacities at the sign-free coupling $J_\pm/J_{zz} = +0.046$, compared with the QMC data of Ref. [4]. Both numerical curves contain the same 2970 self-consistent FCC-32 roots. (b) Dominant heat-capacity peak across the negative-coupling scan. Every point is a paired periodic/masked calculation on the same depth-$d = 1$ microscopic space; the scan stops where either complete 2970-root band ceases to be resolved. The guides show the false torus scale g_4 and physical hexagon scale g_6.

odic) and 2.79 (masked), approaching the expected second and third orders without a fitted subtraction. All 19 scanned negative-coupling bands pass the paired root-count gate through $|J_\pm|/J_{zz} = 0.50$. At that endpoint the dominant periodic and masked peaks are respectively $T/J_{zz} = 4.87 \times 10^{-2}$ and 9.58×10^{-3}; neither value is extrapolated through a partially resolved band.

B. Longitudinal response at every momentum

The same construction gives eigenvectors for dynamical correlations. Let P_α project onto the ice states in transport sector α. The bordered Hermitian matrix

$$\mathcal{H}_\alpha = \begin{pmatrix} P_\alpha H P_\alpha & P_\alpha H Q \\ Q H P_\alpha & Q H Q \end{pmatrix} \tag{8}$$

has the α block of $\Delta F(z)$ as its Schur complement. It is a linear representation of the same masked theory, not a second correction scheme.

For wave vector $\boldsymbol{q}$ and pyrochlore sublattice $\mu = 0, 1, 2, 3$, define

$$S_\mu^z(\boldsymbol{q}) = \sum_{i\in\mu} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i} S_i^z. \tag{9}$$

This operator is diagonal in the Ising basis, so it is evaluated directly between bordered eigenvectors. At zero temperature,

$$S^{zz}(\boldsymbol{q},\omega) = \frac{1}{N}\sum_{\mu,n} |\langle n|S_\mu^z(\boldsymbol{q})|0\rangle|^2 \delta(\omega - E_n + E_0), \tag{10}$$

where ω is energy transfer, $|0\rangle$ has energy E_0, $|n\rangle$ has energy E_n, and the final factor is a Dirac delta.

Figure 2 compares the masked calculation directly with naive periodic diagonalization of the same 140,442-state space. At $J_\pm/J_{zz} = -0.20$, the periodic integrated inelastic weight at $\boldsymbol{q} = 0$ is 0.3126; the masked value is 1.42×10^{-29}. The seven nonzero momenta carry masked weights 0.214–0.299. A repeat at -0.05 gives 0.4879 periodically and 1.36×10^{-30} after masking at $\boldsymbol{q} = 0$, with masked nonzero-momentum weights 0.238–0.309.

The zero follows from transport superselection. The four $\boldsymbol{q} = 0$ sublattice magnetizations are constant within each of the 85 FCC-32 transport sectors, so the projected longitudinal probe cannot connect different eigenstates of a masked ice block. Periodic dynamics mixes those sectors and converts the torus artifact into observable spectral weight. The response comparison is therefore a second, independent benchmark of what the mask removes.

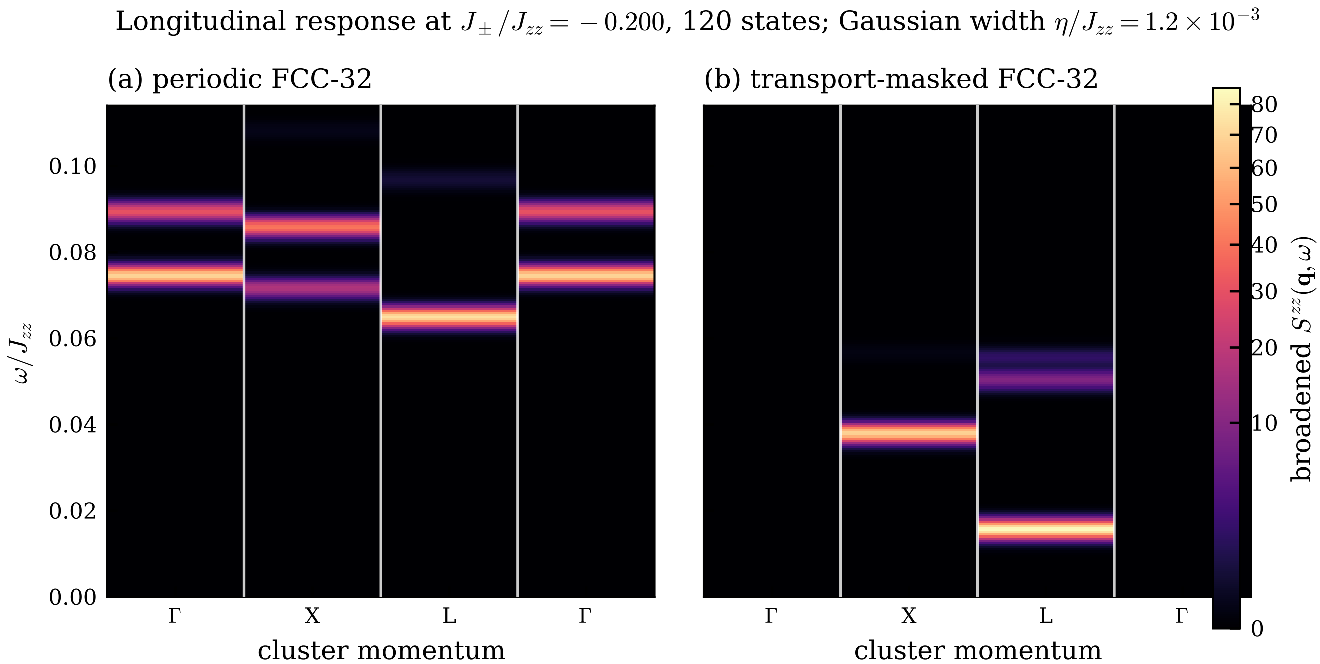

FIG. 2. Longitudinal dynamical structure factor along Γ–X–L–Γ at $J_\pm/J_{zz} = -0.20$. The eight FCC-32 momenta form three high-symmetry stars – one Γ, three X, four L – and each block is one star, averaged over its symmetry-equivalent members; the weights within a star agree to 3×10^{-14}, so the average discards nothing. Blocks are deliberately not interpolated: these are the only momenta the cluster resolves, and both nonzero stars lie on the zone boundary, so no small-$\boldsymbol{q}$ information exists to disperse through. Each block carries a single dominant excitation (88–100% of the weight), consistent with the two transverse photon polarizations being degenerate at both X and L. (a) Naive periodic dynamics and (b) the transport-masked theory use the same depth-$d = 1$ space, 120 final states, four sublattice probes, Gaussian width $\eta/J_{zz} = 1.2\times10^{-3}$, and common color scale. Periodic dynamics produces strong inelastic response at Γ and a higher apparent excitation scale. The mask removes the Γ response while retaining finite weight at both nonzero stars. The removed weight is not in the $U(1)$-protected channel: $\sum_i S_i^z$ commutes with H, so the total longitudinal probe gives exactly zero at Γ in both theories ($< 10^{-31}$), and the periodic Γ intensity resides entirely in the three sublattice-staggered $\boldsymbol{q} = 0$ channels, which nothing protects.

C. What the longitudinal probe can and cannot see

The eight FCC-32 momenta are not eight independent points: they are three high-symmetry stars of the fcc Brillouin zone – one Γ, three X, four L – and both nonzero stars lie on the zone boundary. The cluster therefore carries no small-$\boldsymbol{q}$ information at all, and no statement about a photon *dispersion* can be made on it. What can be made, and is the subject of this subsection, is a statement about which excitations the longitudinal probe is able to reach.

To separate the probe from the spectrum we resolve the eigenvalues by momentum directly, projecting each eigenvector of the masked block onto the cluster momenta with $P_{\boldsymbol{q}} = N_t^{-1}\sum_t \chi_{\boldsymbol{q}}(t)^* T_t$, where T_t runs over the lattice translations. The projector is exact to 4×10^{-13}. Each level then carries a degeneracy equal to its star size times two.

The result is that S^{zz} does not couple to the lowest excitation at X. At $J_\pm/J_{zz} = -0.20$ the lowest X level lies at $\omega/J_{zz} = 0.025274$ and carries longitudinal weight 3.2×10^{-27}, while the weight the structure factor does show at X sits on a higher level at 0.037928. The same holds at $J_\pm/J_{zz} = +0.046$: the lowest X level at 0.003612 carries 4.6×10^{-27}, and the visible weight sits at 0.004525. At L, by contrast, the probe sits on the lowest level and takes 88–90% of that star's weight. Resolving the four sublattice channels separately does not change this: every level the probe reaches has all four channels exactly equal, and every level it misses is at machine zero in all four. The miss is a selection rule, not a form factor.

The rule is spin-flip parity. The Hamiltonian is invariant under the global flip $\mathcal{P} : S_i^z \mapsto -S_i^z$, $S_i^\pm \mapsto S_i^\mp$, which preserves the J_{zz} term trivially and maps the $J_\pm$ term onto its Hermitian conjugate. The longitudinal probe is odd under it, while the hexagon ring exchange – the lattice magnetic flux $\Phi_h = S^+S^-S^+S^-S^+S^- + \text{h.c.}$ – is even:

$$\mathcal{P}\, S_\mu^z(\boldsymbol{q})\, \mathcal{P}^{-1} = -S_\mu^z(\boldsymbol{q}), \qquad \mathcal{P}\, \Phi_h\, \mathcal{P}^{-1} = +\Phi_h. \quad (11)$$

Since the ground state is a parity eigenstate, $S^z(\boldsymbol{q})|0\rangle$ and $\Phi(\boldsymbol{q})|0\rangle$ lie in opposite parity sectors and can reach no level in common.

Figure 3 confirms this directly. Alongside S^{zz} we eval-

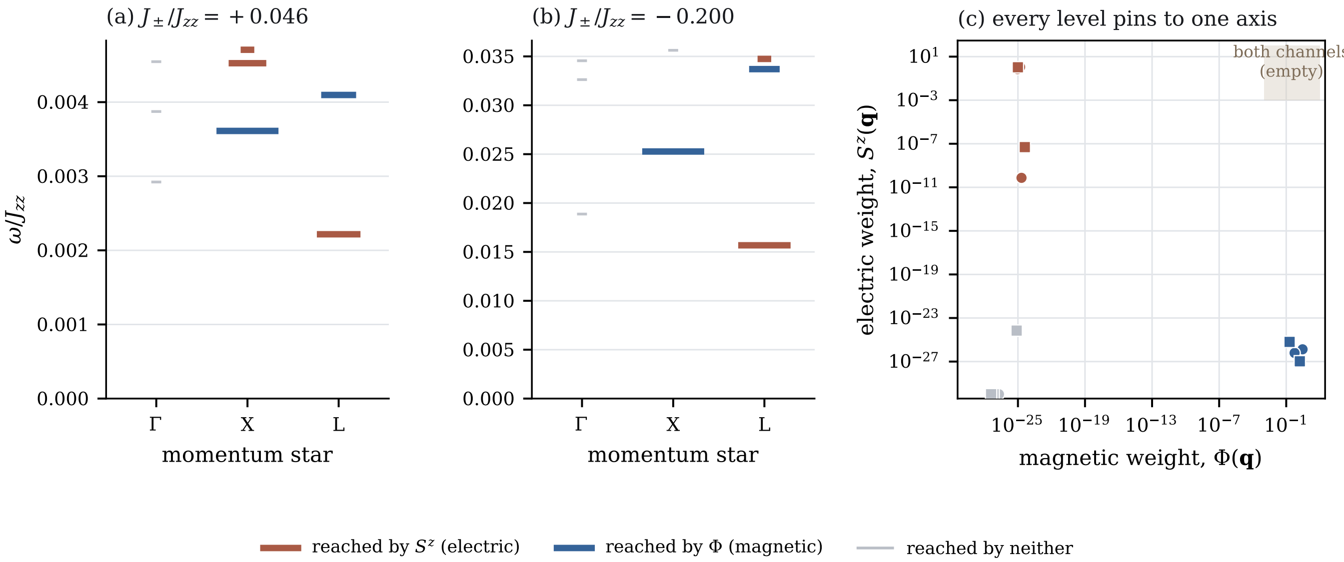

FIG. 3. Electric and magnetic probes reach disjoint sets of levels. (a), (b) Every computed level of the masked block, drawn at its own energy in the column of its momentum star, coloured by which probe reaches it and with bar width encoding the weight; levels reached by neither probe are shown as hairlines so that they remain visible. (c) The electric weight of each level against its magnetic weight. Points pin to one axis or the other over more than twenty-five decades and the diagonal region, where a level reached by both probes would sit, is empty. This is Eq. (11) at work rather than a numerical coincidence.

uate the magnetic channel $|\langle n|\Phi(\boldsymbol{q})|0\rangle|^2$ with $\Phi(\boldsymbol{q}) = \sum_h e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_h}\Phi_h$ summed over the 32 contractible plaquettes. Every level is bright in exactly one channel, with the complement between 10^{-25} and 10^{-35}; at $J_\pm/J_{zz} = -0.20$ the lowest X level carries magnetic weight 1.645 against longitudinal weight 1.1×10^{-27}, and at $+0.046$ the figures are 2.891 and 1.3×10^{-26}. The excitation the longitudinal probe misses is therefore an ordinary gauge excitation seen plainly in the magnetic channel, not a missing or anomalous state.

Two consequences follow for how Fig. 2 should be read. First, the X block of the longitudinal response is not the lowest excitation at X and must not be quoted as one. Second, because no level is reached by both probes, no single level on this cluster satisfies the criterion of carrying both an electric and a magnetic component, so we make no photon assignment here. We note that only the 32 contractible hexagons enter Φ: three quarters of the six-cycles on this cluster wrap the periodic box, and their ring exchange transports polarization between transport sectors, which is precisely the winding artifact the mask removes. Restricted to the contractible set the ring exchange conserves the sector exactly, which is an independent confirmation of the construction of Sec. III.

Finally, none of these energies is converged in system size. Between cubic-16 and FCC-32 the lowest excitation at X moves by a factor 1.5–2.2 and at Γ by 3.5–6.1, and the ordering of the Γ and X levels inverts. The selection rule (11) is exact and size-independent; the energies it acts on are not, and we quote them only as the spectrum of the finite cluster.

V. OPERATOR-RESOLVED LOOP HIERARCHY

Each surviving off-diagonal matrix element $[\Delta F]_{ab}$ flips the sites on which ice configurations a and b differ. This set is a *simple loop* when its nearest-neighbour induced graph is connected and every vertex has degree two. We define

> *L is the number of flipped spins in this set. For a simple loop it is also the number of bonds in the closed path, i.e., its perimeter.*

Thus $L = 6$ is an elementary pyrochlore hexagon. Vertex-joined loops and disconnected products are excluded, preventing products of ordinary rings from being counted as new elementary long loops.

For every even perimeter L, the measured amplitude is

$$K_L = \operatorname*{mean}_{(a,b)\in\mathcal{S}_L} |[\Delta F(z_0)]_{ab}|\,, \tag{12}$$

where $\mathcal{S}_L$ is the set of mask-surviving simple-loop elements and z_0 is the ground energy of the same depth-truncated microscopic Hamiltonian used in the resolvent. Deleted winding entries are absent operators and are not averaged as zeros.

A. Perturbative ratio

The leading coefficient is analytic for any alternating simple loop. Write $L = 2n$, so the loop is completed by n bond exchanges. There are two alternating perfect matchings. At an intermediate step, the chosen bonds form c connected segments and carry c spinon pairs with excitation energy cJ_{zz}. Summing the denominators over every exchange ordering gives

$$K_L^{(0)} = 2\binom{2n-2}{n-1}\frac{|J_\pm|^n}{J_{zz}^{n-1}} = 2\binom{L-2}{L/2-1}\frac{|J_\pm|^{L/2}}{J_{zz}^{L/2-1}}. \quad (13)$$

For $L = 6, 8, 10, 12$, the coefficients are 12, 40, 140, and 504. Defining $x = |J_\pm|/J_{zz}$ gives

$$\frac{K_8^{(0)}}{K_6^{(0)}} = \frac{10}{3}x, \quad \frac{K_{10}^{(0)}}{K_6^{(0)}} = \frac{35}{3}x^2, \quad \frac{K_{12}^{(0)}}{K_6^{(0)}} = 42x^3, \quad (14)$$

or $K_{L+2}^{(0)}/K_L^{(0)} = (4 - 4/L)x$.

Figure 4 is the central operator result. At weak coupling the ratios begin near their expected powers, validating the extraction. With increasing x, all three bend downward relative to Eq. (14). The $d = 2$ results flatten by approximately $x = 0.15$. Expanding the virtual space to $d = 3$ raises their absolute values but preserves their wide separation.

At $x = 0.30$, leading perturbation theory predicts $(K_8, K_{10}, K_{12})/K_6 = (1.00, 1.05, 1.134)$. The matched $d = 3$ calculation instead gives $(0.262, 0.0617, 0.0204)$. These amplitudes are smaller than the leading extrapolation by factors 3.8, 17, and 56. The eight-site loop is about one quarter of the hexagon, the ten-site loop about six percent, and the twelve-site loop about two percent. None approaches dominance.

The data therefore show corrections to a hexagon-only Hamiltonian but not a proliferation of elementary long loops. Repeated virtual excursions renormalize the bare series in a strongly perimeter-dependent way and retain a compact hierarchy. The $d = 3$ calculations are matched 512-column samples, not a complete FCC-32 complement, so the quoted ratios remain finite-volume, depth-controlled estimates. Their separation is nevertheless much larger than the observed depth drift.

VI. EXPERIMENTAL MEANING

The quantities K_L are effective-Hamiltonian coefficients multiplying alternating loop-flip operators

$$W_{\mathcal{C}} = S_1^+ S_2^- S_3^+ \cdots S_L^-, \quad (15)$$

where $\mathcal{C}$ is a closed path of perimeter L. In gauge language, $W_{\mathcal{C}} + W_{\mathcal{C}}^\dagger$ is a magnetic-loop operator. Its expectation value $\langle W_{\mathcal{C}} \rangle$ is related to, but not identical with, the coefficient K_L; measuring one does not uniquely determine the other.

Neutron scattering measures two-spin correlations such as $S^{zz}(\boldsymbol{q}, \omega)$ rather than the equal-time expectation value of the many-spin operator in Eq. (15). The bulk-magnet connection is therefore indirect. Pinch-point evolution and photon weight diagnose emergent electrodynamics [3]; translation fractionalization of zero- and π-flux states produces distinct inelastic spectra [5]; and neutron spectroscopy has resolved a multi-peak continuum consistent with fractional matter coupled to a π-flux gauge background [6]. Figure 2 shows how transport artifacts can enter exactly this experimentally accessible channel.

Direct Wilson-loop measurements are more natural in programmable gauge simulators. Proposed superconducting architectures couple loop operators nondestructively to a cavity [7], and platform-independent protocols reconstruct Wilson loops using local operations [8]. There, perimeter versus area behavior of $\langle W_{\mathcal{C}} \rangle$ diagnoses deconfinement, while controlled loop perturbations can spectroscopically constrain the coefficients K_L.

VII. CONCLUSION

Every finite-cluster calculation in this work is made on FCC-32. This removes cluster substitution as a hidden variable: periodic and masked thermodynamics use the same 2970 roots, periodic and masked responses use the same 140,442-state space and all eight momenta, and loop amplitudes are extracted from the same FCC-32 geometry. The sole correction is transport masking of the Feshbach map; no counterterm is introduced.

The results support two conclusions. First, the finite torus generates a false second-order gauge scale and large spurious longitudinal response at $\boldsymbol{q} = 0$. Removing transport-changing matrix elements restores the contractible dynamics while retaining every nonzero FCC-32 momentum. Second, the recovered dynamics remains local in loop perimeter far beyond what bare order counting suggests. At $|J_\pm|/J_{zz} = 0.30$, longer loops that perturbation theory predicts to rival the hexagon remain smaller by factors from four to more than fifty.

The calculation is still finite-volume and depth truncated. It does not establish a thermodynamic-limit photon pole or phase boundary, and the 2970-state heat capacity is only the low-energy ice contribution. Larger hosts and deeper complete complements are required for those claims. Within these limits, the FCC-32 comparisons isolate a sharp result: transport masking removes an observable torus artifact, and the nonperturbative theory suppresses rather than accelerates the proliferation of long Wilson loops.

[1] M. Hermele, M. P. A. Fisher, and L. Balents, Phys. Rev. B **69**, 064404 (2004).

[2] M. J. P. Gingras and P. A. McClarty, Rep. Prog. Phys.

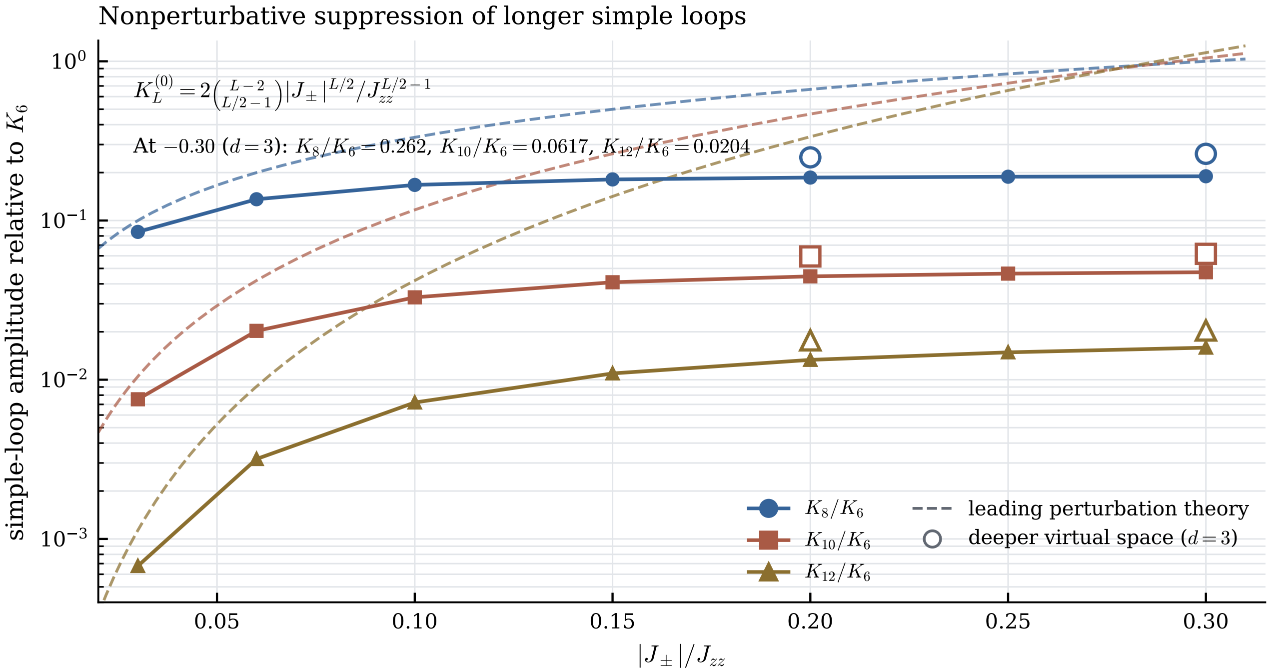

FIG. 4. FCC-32 simple-loop hierarchy. Solid symbols are the complete virtual-depth-$d=2$ results; open symbols are matched virtual-depth-$d=3$ checks. Dashed curves are the leading ratios in Eq. (14). Bare perturbation theory predicts that the longer loops approach or exceed the hexagon near $|J_\pm|/J_{zz} = 0.30$; the nonperturbative Feshbach amplitudes flatten far below it.

77, 056501 (2014).

[3] O. Benton, O. Sikora, and N. Shannon, Phys. Rev. B **86**, 075154 (2012).

[4] C.-J. Huang, Y. Deng, Y. Wan, and Z. Y. Meng, Phys. Rev. Lett. **120**, 167202 (2018), arXiv:1707.00099.

[5] F. Desrochers, L. E. Chern, and Y. B. Kim, Phys. Rev. B **107**, 064404 (2023).

[6] V. Porée, H. Yan, F. Desrochers, S. Petit, E. Lhotel, M. Appel, J. Ollivier, Y. B. Kim, A. H. Nevidomskyy, and R. Sibille, Nat. Phys. **21**, 83 (2025).

[7] G. K. Brennen, G. Pupillo, E. Rico, T. M. Stace, and D. Vodola, Phys. Rev. Lett. **117**, 240504 (2016), arXiv:1512.06565.

[8] E. Zohar, Phys. Rev. D **101**, 034518 (2020), arXiv:1911.11156.